

SOF IMO — Topic Deep-Dive: Geometry & Data Handling

For Ambar, Class III CBSE

PART A — 2D SHAPES (flat shapes)

A flat shape has **sides** (straight edges) and **corners** (where two sides meet).

Shape	Sides	Corners	Remember
Triangle	3	3	tri = three
Square	4	4	ALL 4 sides equal
Rectangle	4	4	opposite sides equal (2 pairs)
Pentagon	5	5	penta = five
Hexagon	6	6	hexa = six
Octagon	8	8	octa = eight (like octopus)
Circle	0	0	no straight side, no corner

Square vs rectangle trap: both have 4 sides and 4 corners; only the square has ALL sides equal.

Stretch idea — perimeter: the distance all around a shape. A square of side 6 cm has perimeter $6 + 6 + 6 + 6 = 24$ cm.

PART B — 3D SHAPES (solid shapes)

Solid shapes have **faces** (flat surfaces), **edges** (where two faces meet) and **corners/vertices** (points).

Solid	Faces	Edges	Corners	Everyday example
Cube	6	12	8	dice
Cuboid	6	12	8	brick, matchbox, book
Sphere	0 flat (1 curved surface)	0	0	ball, football
Cylinder	2 flat + 1 curved	0 straight	0	tin can, gas cylinder
Cone	1 flat + 1 curved	0 straight	1 point (apex)	ice-cream cone, joker's cap

Rolling and sliding: **sphere** only rolls; **cube/cuboid** only slide; **cylinder and cone** can roll AND slide (they have both curved and flat surfaces).

2D vs 3D trap: circle, square, triangle are FLAT (2D); sphere, cube, cone are SOLID (3D). A dice is a cube, not a square.

PART C — PATTERNS

- Repeating patterns: circle, square, circle, square, ____ → circle. Find the repeating block first.
- Growing patterns: 2, 4, 6, 8 (+2); 5, 10, 15 (+5); A, AB, ABC (one letter added).
- Backward patterns: Z, Y, X, W → V; 60, 55, 50 → 45.
- Stretch: 1, 4, 9, 16, 25 (square numbers: 1×1 , 2×2 , 3×3 , ...).

PART D — DATA HANDLING

Pictographs

A pictograph uses symbols. **Always read the key first** — one symbol may stand for MORE than one thing.

Worked example: Key: each symbol = 2 children. Mango row has 5 symbols → $5 \times 2 = 10$ children like mango.

Comparison example: each symbol = 5 trees; Class 3 has 4 symbols (20 trees), Class 4 has 6 symbols (30 trees) → Class 4 planted $30 - 20 = 10$ more.

Tally charts

Tally marks are counted in bundles of 5: four lines with the fifth across (||||).

- |||| = 4, |||| (crossed) = 5, |||| || = 7, |||| |||| = 10.

Reading tables

“Most/fewest votes” → compare the numbers. “How many in all” → ADD every row (Red 8 + Blue 6 + Green 4 = 18).

Common traps

Trap	Wrong	Right
Ignoring the pictograph key	5 symbols = 5 children	$5 \times 2 = 10$ (key says 2 each)
Cube faces vs corners	cube has 8 faces	6 faces, 8 corners, 12 edges
Square = rectangle mixed up	rectangle has all sides equal	only square does
Circle “has 1 side”	1	0 straight sides, 0 corners
“How many more” answered with total	adds the two rows	subtract the two rows
Cone corners	0	1 point (apex)

High-frequency IMO question types

1. Sides/corners of a named 2D shape; odd one out (2D among 3D).
2. Faces/edges/corners of cube, cuboid; everyday-object shape matching.
3. Which solid rolls / slides / both.
4. What comes next in a shape, number or letter pattern.
5. Pictograph with a key: value of a row, difference between rows.
6. Tally count; most/fewest from a small table; total votes.

Key facts checklist

- ☐ Triangle 3, square/rectangle 4, pentagon 5, hexagon 6, octagon 8, circle 0
- ☐ Cube and cuboid: 6 faces, 12 edges, 8 corners
- ☐ Sphere: no flat face, no edge, no corner
- ☐ Cylinder: 2 flat faces; cone: 1 flat face + 1 point
- ☐ Read the pictograph KEY before counting
- ☐ Tally |||| (crossed) = 5

Quick drill (10 questions)

1. How many corners does a rectangle have?
2. How many faces does a cube have?
3. Which solid has no corners and no edges at all?
4. What comes next: 3, 6, 9, 12, ____?
5. A cone has how many flat faces?
6. Pictograph key: 1 symbol = 4 books. A row shows 3 symbols. How many books?
7. Tally marks |||| || stand for?
8. Which shape has 6 sides?
9. Votes: Ludo 9, Chess 4, Carrom 6. Which got the most?
10. What comes next: square, circle, circle, square, circle, circle, ____?

Answers: 1) 4 2) 6 3) sphere 4) 15 5) 1 6) 12 7) 8 8) hexagon 9) Ludo 10) square